

BACHELOR OF COMPUTER APPLICATION (BCA)

YEAR: THIRD, SEMESTER-VI

(To be effective from the session 2025-2026)

S. No.	Paper Code	Subject	Periods			Evaluation Scheme				Sub Total	Credit
			L	T	P	Sessional Exam			Exam ESE		
						CT	TA	Total			
1.	NBCA-601	Machine Learning	3	1	0	20	10	30	70	100	4
2.	NBCA-602	Multimedia System	3	1	0	20	10	30	70	100	4
3.	NBCA-603	Software Project Management	3	0	0	20	10	30	70	100	3
4.	NBCA-604X	Departmental Elective-II	3	0	0	20	10	30	70	100	3
Practical											
5.	NBCA-605P	Machine Learning Lab	0	0	3		20	20	30	50	2
6.	NBCA-606P	Seminar	0	0	3		20	20	30	50	2
7.	NBCA-607P	Project Phase-II	0	0	9		50	50	100	150	6
8.	NBCA-GP	General Proficiency	-	-	-		-	--	-	50	
		Total	12	2	15					650	24

Departmental Elective-II

NBCA-6041	Open Source Software
NBCA-6042	Mobile Computing
NBCA-6043	Cryptography
NBCA-6044	Cyber Forensic Analytics

Note: If the student leaves the Programme after completing Semester-VI successfully, student will be awarded a **Bachelor Degree in Computer Application.**

MACHINE LEARNING

L	T	P
3	1	0

Course Outcomes (COs):

After the completion of this course, students will be able to

- Explain the fundamental usage of the concept Machine Learning system
- Demonstrate on various regression Technique
- Analyse the Ensemble Learning Methods
- Illustrate the Clustering Techniques and Dimensionality Reduction Models in Machine Learning.
- Discuss the Neural Network Models and Fundamentals concepts of Deep Learning

Unit-I

8

Introduction: Artificial Intelligence, Machine Learning, Types of Machine Learning Systems, Challenges of Machine Learning. Statistical Learning: Introduction, Supervised and Unsupervised Learning, Training and Test Loss, Tradeoffs in Statistical Learning, Estimating Risk Statistics, Sampling distribution of an estimator.

Unit-II

8

Supervised Learning: Basic Methods: Distance based Methods, Nearest Neighbours, Decision Trees, Naive Bayes, Linear Models: Linear Regression, Logistic Regression, Generalized Linear Models, Support Vector Machines, Binary Classification.

Unit-III

8

Ensemble Learning: Introduction, Voting Classifiers, Bagging and Pasting, Random Forests, Boosting, Stacking. Support Vector Machine: Linear SVM Classification, Nonlinear SVM Classification SVM Regression, and Naïve Bayes Classifiers.

Unit-IV

8

Unsupervised Learning: Clustering, K-Means, Limits of K-Means, Using Clustering for Image Segmentation, Using Clustering for Preprocessing, Using Clustering for Semi-Supervised Learning, DBSCAN, Gaussian Mixtures. Dimensionality Reduction: The Curse of Dimensionality, Main Approaches for Dimensionality Reduction, and PCA.

Unit-V

8

Neural Networks: Introduction to Artificial Neural Networks with Keras, Implementing MLPs with Keras, TensorFlow, Loading and Preprocessing Data with TensorFlow.

Text Books:

1. Alpaydin, E. 'Introduction to machine learning', MIT press.
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition, O'Reilly Publications, 2019
3. Data Science and Machine Learning Mathematical and Statistical Methods, Dirk P. Kroese, Zdravko I. Botev, Thomas Taimre, Radislav Vaisman, 25th November 2020

Reference Books:

1. Mitchell, T.M. 'Machine Learning', McGraw-Hill
2. Ian Goodfellow, Yoshua Bengio and Aaron Courville, 'Deep Learning', MIT Press.
3. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani 'An Introduction to Statistical Learning', Springer.
4. Machine Learning Probabilistic Approach, Kevin P. Murphy, MIT Press, 2012.

MACHINE LEARNING LAB

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Course Outcomes (COs):

At the end of this course students will be able to:

- Implement procedures for the machine learning algorithms
- Design and develop python programs for various learning algorithms
- Apply appropriate data sets to the machine learning algorithms
- Develop machine learning algorithms to solve real world problems

LIST OF PRACTICALS

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate- Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
3. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
4. Exercises to solve the real-world problems using the following machine learning methods: a) Linear Regression b) Logistic Regression c) Binary Classifier
5. Develop a program for Bias, Variance, Remove duplicates , Cross Validation
6. Write a program to implement Categorical Encoding, One-hot Encoding
7. Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
8. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions.
9. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.
10. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.
11. Apply EM algorithm to cluster a Heart Disease Data Set. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.
12. Exploratory Data Analysis for Classification using Pandas or Matplotlib.

13. Write a Python program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set
14. Write a program to Implement Support Vector Machines and Principle Component Analysis
15. Write a program to Implement Principle Component Analysis

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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